25901925 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

Third SEMESTER

Value Addition Course (VAC) - MG3VACSTA200 - APPLIED STATISTICAL ANALYSIS : ETHICAL DATA COLLECTION, INTERPRETATION AND DECISION MAKING IN SOCIETY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type questions

Answer any **7** questions

Each question carries **2** marks

1. Define tabulation of data. [K] / [CO1]

2. What is the difference between simple and multiple linear regression? [K] / [CO1]

3. A set of class frequencies that completely specifies the population is called a _____ set. [K] / [CO1]

4. Give any two merits of Standard deviation. [K] / [CO1]

5. A school claims that its students score an average of 82 on a standardized test. A random sample of 20 students is taken. The sample's average score is 84 with a [K] / [CO2]

standard deviation of 8.1.Obtain the test statistic value for testing the one sample mean t test.

6. Write the test statistic for testing a population proportion in one sample. [U] / [CO2]

7. What is Type II error in hypothesis testing? [K] / [CO2]

8. In linear regression, what is the line called that best fits the data? [U] / [CO1]

9. Which chart is used to represent data divided into rectangular bars? [U] / [CO1]

10. What is the name of the primary R function used to perform a Chi-square test for independence? [An] / [CO3]

[2x7 = 14]

Part B

Short Essay Type Questions

Answer any **4** questions

Each question carries **6** marks

11. If a sample of size 22 items has a mean of 15 and another sample of size 18 items has a mean of 20. Find the mean of the combined sample. [A] / [CO1]

12. Draw a histogram for the following data. [U] / [CO1]

0–10	10–20	20–30	30–40	40–50
5	8	10	7	5

13. Differentiate between Primary Data and Secondary Data with examples. [K] / [CO1]

14. A teacher wants to know whether a new teaching method is effective. She takes the marks of **6 students** before and after using the new method. [A] / [CO2]

Student	Before	After
1	45	49
2	50	52
3	43	39
4	41	42
5	42	38
6	44	47

Test at 5% significance level, whether the new method improves the performance.

15. The average height of students in a college is claimed to be 160 cm. A random sample of 64 students gives a mean height of 162 cm with a standard deviation of 8 cm. Test at 5% level of significance whether the claim is correct. [A] / [CO2]

16. Write the null and alternative hypotheses for testing whether the population mean equals a specific value. [U] / [CO3]

[6x4 = 24]

Part C

Essay type questions

Answer any **1** questions

Each question carries **12** marks

17. A study examines the association between a father's eye color and his son's eye color, providing the following data: (Father Dark eyes, Son Dark eyes) = 50 (Father Dark eyes, Son Brown eyes) = 89 (Father Brown eyes, Son Dark eyes) = 79 (Father Brown eyes, Son Brown eyes) = 782 [U] / [CO1]

    Let A denote the attribute 'Father has Dark eyes' and B denote 'Son has Dark eyes'.

    (a) Write down the given data using the standard notation for ultimate class frequencies (e.g., (AB), (Aβ), etc.).

    (b) Calculate Yule's Coefficient of Association (Q) based on this data, showing your formula and calculation.

    (c) Based on the calculated value of Q, interpret the nature and intensity of the association between the eye color of fathers and sons.

18. (a). Explain the method of testing the significance of the difference between two means in large samples. [A] / [CO2]

    (b).Two brands of soap were tested for durability. A sample of 50 cakes of Brand X lasted an average of 18 days with a standard deviation of 2.4 days, while a sample of 60 cakes of Brand Y lasted an average of 17.5 days with a standard deviation of 2.8 days. Test whether there is a significant difference in the average life of the two brands at 5% level of significance.

[12x1 = 12]